

Question 1

- What are the two categories of data compression methods?
- *The two categories of data compression methods are lossless and lossy.*

Question 2

- What is the difference between lossless compression and lossy compression?
- *Lossless compression preserves the integrity of the data; in lossy compression, some of the data is lost in the compression/decompression process.*

Question 3

- What is run-length encoding?
- *Run length encoding is a lossless compression method in which repeated occurrences of a symbol are replaced by one occurrence of the symbol followed by the number of occurrences.*

Question 4

- How does Lempel Ziv encoding reduce the amount of bits transmitted?
- *Lempel-Ziv (LZ) encoding is a type of dictionary-based, lossless compression method in which a dictionary is constructed during encoding.*
- *During decoding, the dictionary is reconstructed from the received data.*
- *During encoding or decoding, the already-encountered strings can be substituted by their index in the dictionary to reduce the amount of information transmitted.*

Question 5

- What is Huffman coding?
- *Huffman coding uses the frequency of the characters in the file to construct a tree. The tree is then used to generate codes for each character with the more frequent characters having shorter codes than the less frequent characters.*

Question 6

- What is the role of the dictionary in LZ encoding?
- *The dictionary in LZ encoding consists of indexed entries that refer to substrings in the original file. These indices are used to refer to future occurrences of these substrings.*

Question 7

- What is the advantage of LZ encoding over Huffman coding?
- *In Huffman coding, both the sender and receiver must have a copy of the same code in order for the decoded file to match the encoded file. In LZ encoding, the dictionary is generated from the data itself.*

Question 8

- Name three lossy compression methods.
- *The three lossy compressions are JPEG, MPEG, and MP3.*

Question 9

- Encode the following bit pattern using run-length encoding with 5-bit codes:

18 zeros, 11, 56 zeros, 1, 15 zeros, 11

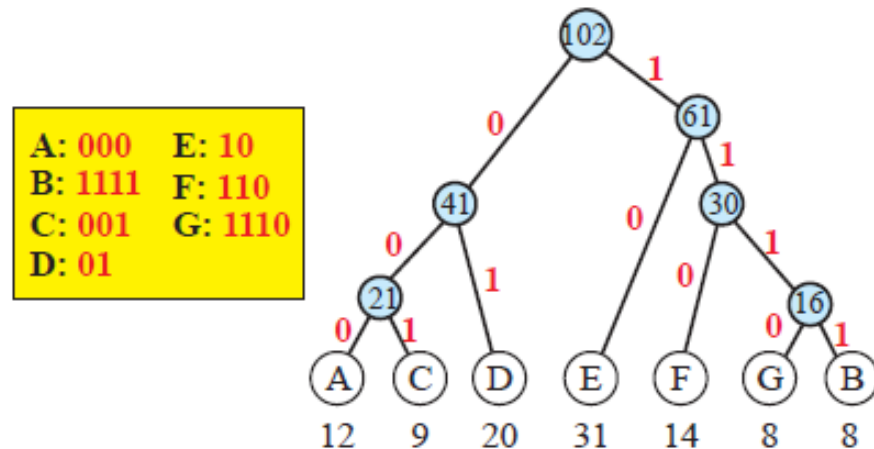
- 10010 00000 11111 11001 01111 00000 00000

Question 10

- Encode the following characters using Huffman coding with the given frequencies:

A (12), B (8), C (9), D (20), E (31), F (14), G (8)

- Different codes result from different ways of organizing the tree. One possible tree with the resulting code is shown Figure*

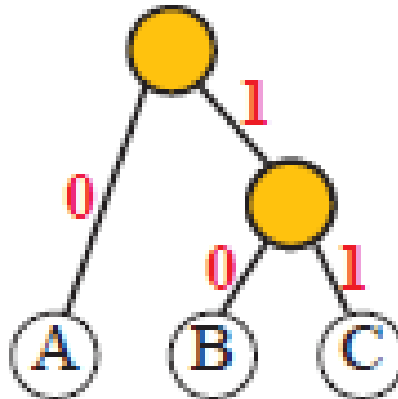


Question 11

- Encode the message BAABBBBAACAA using the following Huffman code:

A: 0 B: 10 C: 11

A: 0
B: 10
C: 11



Question 12

- Decode the message 0101000011110 using the following Huffman code:

A: 0 B: 10 C: 11

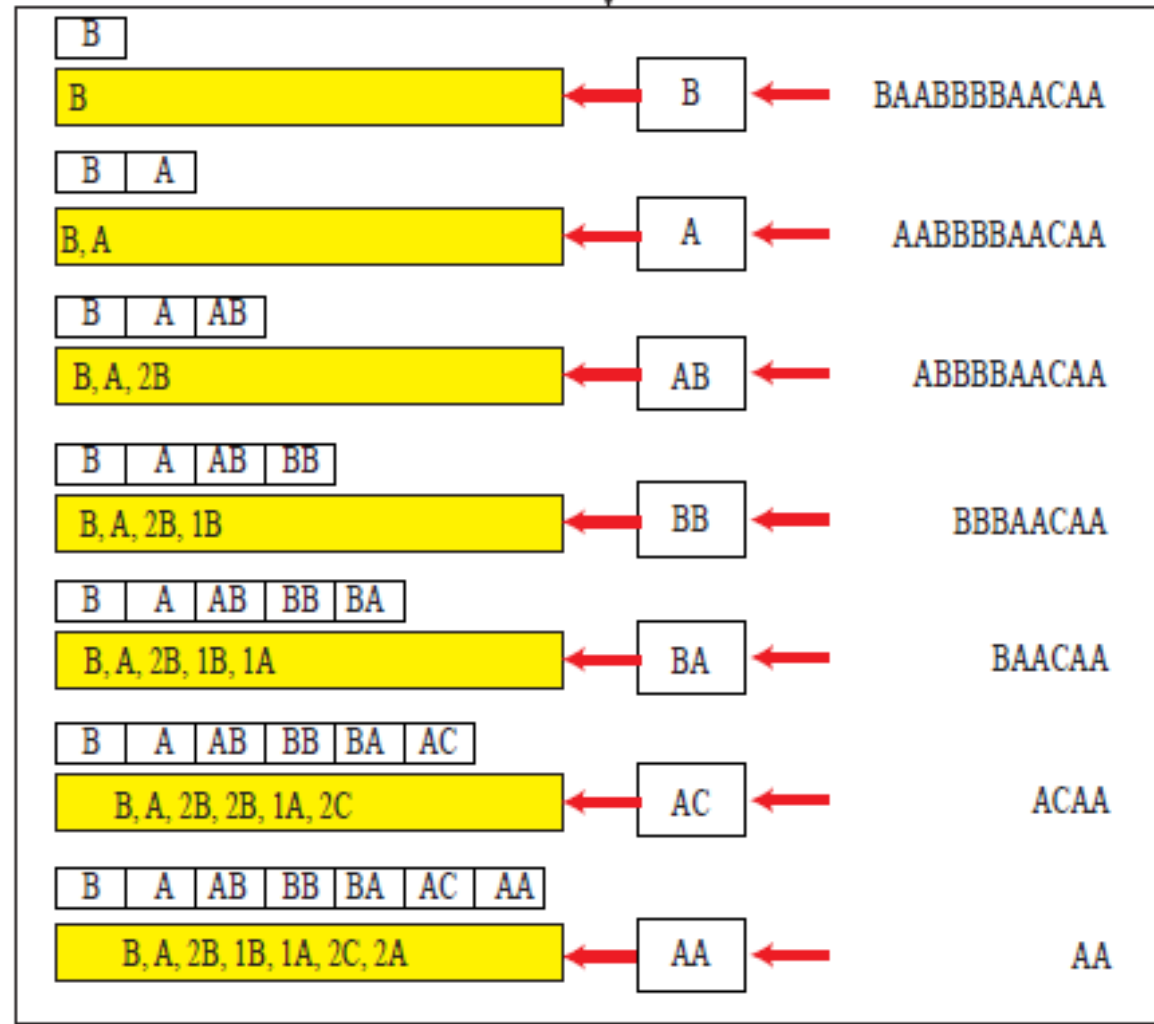
ABBAAACCA

Question 13

- Encode the message BAABBBBAACAA using the Lempel Ziv method, then decode the encoded message to get the original message.

Uncompressed

BAABBBBAACAA

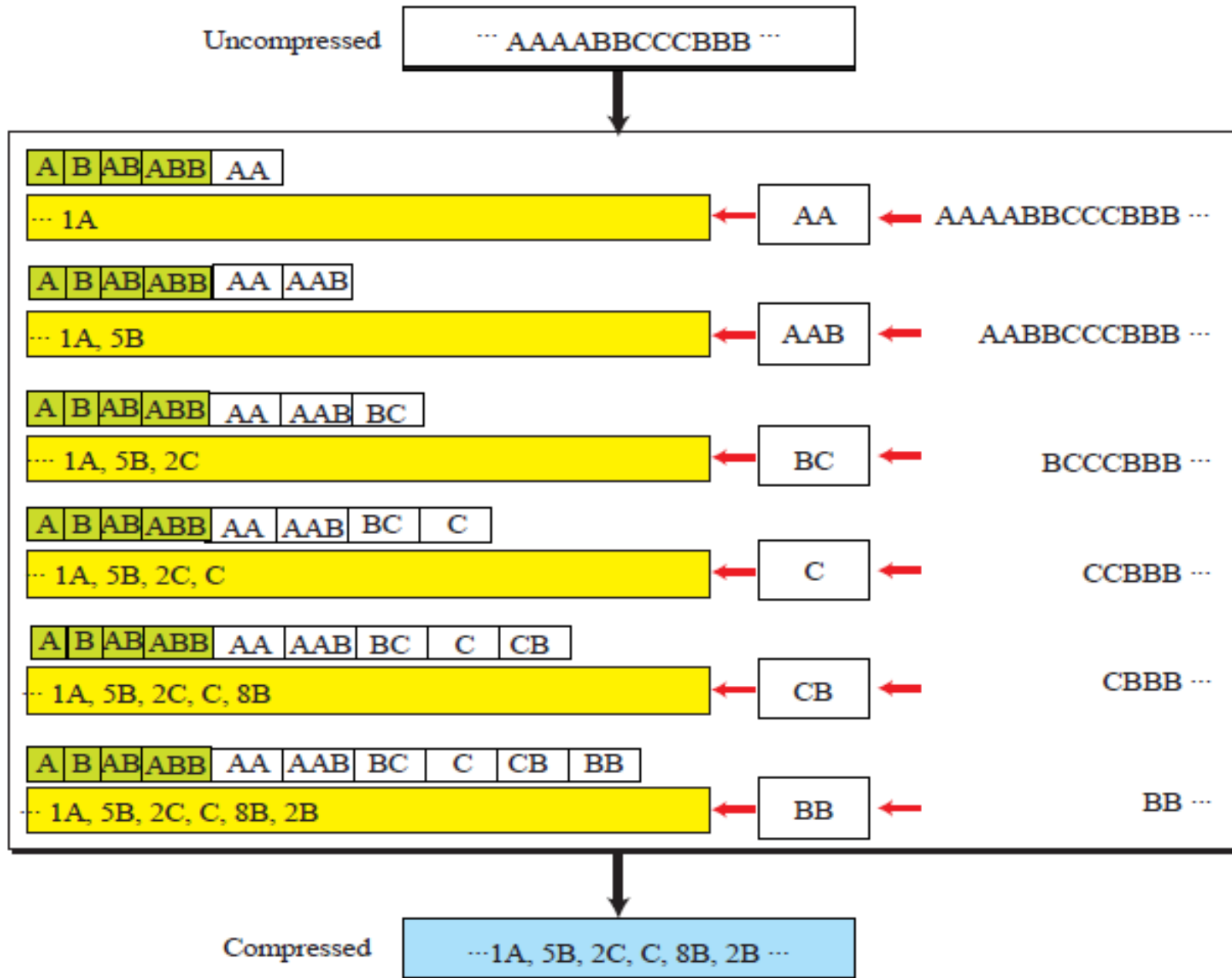


Compressed

B, A, 2B, 1B, 1A, 2C, 2A

Question 14

- Encode the string AAAABBBCCCBBB (part of a message) using the Lempel Ziv method if the dictionary contains ABB. Show the final contents of the dictionary.



if the dictionary contains ABB, it must also have A, B, AB.

Question 15

- Prove the statement stated as under:

"If a receiver knows the message being transmitted, the amount of information carried will be zero."

- Here it is stated that receiver knows the message. This means that only one message is transmitted. Thus, probability of occurrence of this message will be $P(x_i) = 1$. This is because only one message and its occurrence is certain (probability of certain event is '1'). The amount of information carried by this type of message will be,

$$I(x_i) = \log_2 \frac{1}{P(x_i)} = \frac{\log_{10} 1}{\log_{10} 2}$$

$$I(x_i) = 0 \text{ bits}$$

Question 16

- Verify the following expression:

$$I(x_i x_j) = I(x_i) + I(x_j) \text{ if } x_i \text{ and } x_j \text{ are independent}$$

Solution: If x_i and x_j are independent, then we know that

$$P(x_i x_j) = P(x_i)P(x_j)$$

Also
$$I(x_i x_j) = \log \frac{1}{P(x_i x_j)} = \log \frac{1}{P(x_i)P(x_j)} = \log \frac{1}{P(x_i)} + \log \frac{1}{P(x_j)}$$

or
$$I(x_i x_j) = I(x_i) + I(x_j) \quad \textbf{Hence Proved.}$$

Question 17

- Calculate the amount of information if it is given that $P(x_i) = \frac{1}{4}$.
- 2bits

Question 18

- A Discrete Memoryless Source (DMS) X has four symbols x_1, x_2, x_3, x_4 with probabilities
$$P(x_1) = 0.4, P(x_2) = 0.3, P(x_3) = 0.2, P(x_4) = 0.1.$$
 - (i) Calculate $H(X)$.
 - (ii) Find the amount of information contained in the messages $x_1 x_2 x_1 x_3$ and $x_4 x_3 x_3 x_2$, and compare with the $H(X)$ obtained in part (i).

Solution: (i) We know that entropy is given by

$$H(X) = - \sum_{i=1}^4 P(x_i) \log_2 [P(x_i)]$$

Substituting values and simplifying, we get

$$H(X) = - 0.4 \log_2 0.4 - 0.3 \log_2 0.3 - 0.2 \log_2 0.2 - 0.1 \log_2 0.1$$

Solving, we get

$$H(X) = 1.85 \text{ b/symbol} \quad \mathbf{Ans.}$$

(ii) Now, we have $P(x_1 x_2 x_1 x_3) = (0.4)(0.3)(0.4)(0.2) = 0.0096$

Therefore, $I(x_1 x_2 x_1 x_3) = -\log_2 0.0096 = 6.70 \text{ b/symbol}^* \quad \text{Ans.}$

Thus, $I(x_1 x_2 x_1 x_3) < 7.4 [= 4H(X)] \text{ b/symbol}$

Also, $P(x_4 x_3 x_3 x_2) = (0.1)(0.2)^2(0.3) = 0.0012$

Therefore $I(x_4 x_3 x_3 x_2) = -\log_2 0.0012 = 9.70 \text{ b/symbol}$

We conclude that, $I(x_4 x_3 x_3 x_2) > 7.4 [= 4H(X)] \text{ b/symbol} \quad \text{Ans.}$

Question 19

- A DMS X has five symbols x_1, x_2, x_3, x_4 , and x_5 with $P(x_1) = 0.4$, $P(x_2) = 0.19$, $P(x_3) = 0.16$, $P(x_4) = 0.15$, and $P(x_5) = 0.1$.
- Construct a Huffman code for X , and calculate the efficiency of the code.

TABLE 6. 13.

